**LESSON 117 (0.5 GROUND and 1.5 DUAL)**

## Lesson Objective

The purpose of this lesson is to introduce slips, emergency procedures, and non-towered airport operations. Slips are practiced for use in emergency situations, to clear obstacles during the landing approach and crosswind situations. The applicant should be able to make radio calls with little instructor guidance.

## Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
    - Warrior Maneuver's Manual.
        - Non-towered operations.
    - Emergency checklists.
    - V speed and Emergency quiz review.
- Student questions.

## References

- Advisory Circular. (Search Tool)
    - Non-Towered Airport Flight Operations.
- Non-Towered Airport Flight Operations. Airplane Flying Handbook. (FAA-H-8083-3C)
    - Chapter 05 - Maintaining Aircraft Control: Upset Prevention and Recovery Training.
    - Chapter 06 - Takeoffs and Departure Climbs.
    - Chapter 08 - Airport Traffic Patterns.
    - Chapter 09 - Approaches and Landings.
    - Chapter 18 - Emergency Procedures
- Warrior Maneuvers Manual

## Instructor Guidance

- Before departing, quiz the student on the emergency checklists — specifically the engine failure in flight procedure. If they cannot recite the ABCD flow (Airspeed, Best field, Checklists, Declare) from memory, address it on the ground. A student who has to search for the procedure during a simulated emergency in the aircraft is not ready for this lesson.
- When reviewing the V-speed and emergency quiz results from last lesson, use any incorrect items as a focused teaching moment rather than simply providing the correct answers. Ask the student why they think the correct answer is what it is — this builds understanding rather than just correction.
- This lesson introduces emergency procedures in the aircraft for the first time. Set clear expectations during the pre-brief: the student will never be surprised by a simulated engine failure below 500 feet AGL, and the aircraft will never be allowed to descend to an unsafe altitude during any emergency maneuver. Establishing this safety framework helps the student engage with the maneuver confidently rather than anxiously.
- For the engine failure demonstration, strictly follow the telling and doing sequence —

instructor demonstrates with student narrating, then student performs with narration. Pay close attention to the student's choice of landing area. Wire strike avoidance must be emphasized every single time. Florida's low terrain and abundance of power lines make this a real operational hazard.

- When trimming for V-glide, do not allow the student to chase the airspeed indicator. Establish the correct pitch attitude first, then verify airspeed. Students who fixate on the airspeed indicator during an engine failure lose precious seconds of situational awareness.
- For the emergency descent, confirm the student checks the area before initiating the maneuver and monitors airspeed closely throughout. Exceeding VFE with flaps extended is a structural concern — guard against it and address it immediately if it occurs.
- Non-towered airport operations at Valkaria or Sebastian will likely be a new environment for the student. Brief the specific procedures for the destination airport before departure — which runway is typically in use, CTAF frequency, and expected traffic pattern direction. A student who arrives at an unfamiliar non-towered airport unprepared will be overwhelmed.
- During non-towered pattern operations, the student should be making all radio calls independently with you assisting only as needed. If they miss a call or use incorrect phraseology, prompt them to make the corrected call rather than making it for them. The completion standard requires radio calls with little instructor guidance.
- For forward slips, watch closely for the student raising the nose during the slip. A nose-high attitude during a forward slip combined with crossed controls creates a dangerous cross-controlled stall scenario — this must be monitored and corrected without hesitation.
- The go-around decision in this lesson should be treated as a deliberate teaching moment. Brief specific scenarios where a go-around is mandatory — unstabilized approach, incorrect runway, conflicting traffic — and reinforce that the decision to go around should always be made early and without hesitation.
- With Lesson 121 pre-solo tests approaching, remind the student during the post-brief that memorization of emergency checklists and V-speeds is not optional — it is a prerequisite for solo flight.

## <u>Lesson Content (Ground)</u>

### Non-Towered Airport Operations

- Entry:
  - Determine the weather at the field.
  - Determine how to enter the pattern.
    - Enter 45° to the downwind leg.
    - Overfly the field at 1500 ft.
  - Radio calls must be made within 10 NM of the airport.
- Traffic Pattern:
  - Radio calls must be made on every leg.
  - The pattern should be flown at the appropriate traffic pattern altitude.
  - Cross-wind should be turned at 300’ below traffic pattern altitude
- Exiting the Traffic Pattern:
  - When departing the traffic pattern, continue straight out, or exit with a 45°turn in the

direction of the traffic pattern beyond the departure end of the runway. Departures must climb to the traffic pattern altitude before turning.

**Radio Communications in Non-Controlled Airspace**

- Although radio communication in non-controlled airspace are not legally required, it is highly recommended and FITA requires it!
    - Collision avoidance.
- Radio calls must be made within 10 NM, and during every leg of the pattern.
    - Distance can be obtained by going to the nearest page on the G430.
- All radio calls in a non-controlled airspace should be made using the full tail number.
    - Ex: "Valkaria traffic, N602FT is 10 NM north of the airport and will be entering 45 degrees to the left downwind for runway 10, Valkaria traffic."

**Engine Failure in Flight (Emphasis on Pattern)**

- Order of operations
    - A - Airspeed
        - Airspeed comes first and is critical to maintain throughout the maneuver because it guarantees you the most possible time in the air to make a safe landing.
        - Review why we use $V_{Glide}$ (73) and not another airspeed.
            - Total drag curve.
            - Show where 73 is and why going at 63 or 83 (for example) would be worse.
    - B - Best field
        - After you can keep yourself in the air for as long as possible, the next most important consideration is where to put the airplane when that time runs out.
    - C - Checklists
        - After you know you can put the airplane safely on the ground somewhere, the next step to increase the likelihood of a safe outcome is to try and restart the engine. The emergency checklists should be memorized in a flow pattern. If altitude permits, it is also recommended to verify no line items were missed with the physical checklist.
    - D - Declare
        - Aviate, Navigate, Communicate. If safe to do so, declare an emergency on 121.5 and squawk 7700.
        - "mayday mayday mayday (tail number) (location) (emergency) (souls on board) mayday mayday mayday".
        - You can press and hold the frequency toggle soft button to automatically put 121.5 in the active frequency box in COM 1.

**Emergency Approach and Landing**

- This is performed to simulate a situation where power is lost in flight.
- Ensure that the student is familiar with the procedures and has the checklists memorized!
- Quiz the student on the procedures and address any questions they might have.

**Emergency Descent**

- This is performed to simulate a situation in which the airplane needs to descend in the most expeditious means possible.
- Smoke in the cockpit, electrical fire, hypoxic passenger, depressurization, general emergency situations.
- Quiz the student on the procedures for the maneuver and address any questions they might have.

**Slips: Forward/Side**

- Forward:
  - A forward slip is performed to reduce altitude more rapidly without gaining airspeed.
  - This is accomplished using the ailerons to lower the upwind wing while simultaneously using the rudder in the opposite direction.
  - Keeping the nose aimed at the aiming point should be emphasized to the student, as a nose high attitude can result in a cross-controlled stall.
  - Power must be at idle.
  - Anticipate the recovery from the slip in order to resume normal glide path.
- Side:
  - Used to maintain a desired ground track in conditions with wind present.
  - This is accomplished using the ailerons to lower the upwind wing while simultaneously using the rudder in the opposite direction to keep the aircraft lined up with centerline.

**Go/No-Go Decision**

- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

**<u>Lesson Content (Dual)</u>**

**Checklist Procedures**

- Re-emphasize the importance of using checklists.
- Have the student run through all the checklists.
- Student should be able to perform all briefings.

**Preflight Inspection**

- Ensure that the student conducted a thorough preflight inspection.

**Operation of the Communications and Navigation Units**

- Ensure that the student understands how to properly use COMS and NAV units.

**Visual Scanning and Collision Avoidance**

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

**Normal/Crosswind Takeoff**

- The student should conduct the takeoff.
- Ensure the runway entry check and the lineup check are completed.
- Ensure proper crosswind correction is maintained.

**Review Previous Exercises as Required**

- Review any areas the student might need help with by utilizing the pertinent lesson plans.

**Engine Failure in Flight (Emphasis on Pattern)**

- First completion of the maneuver will be a demonstration.
    - <u>Instructor does, student says.</u>
        - Verify correct procedure.
    - Points of Emphasis.
        - Trimming for $V_{glide}$.
        - Wire strike avoidance.
        - Choice of landing point.
- Second completion of the maneuver will be performed by the student.
    - <u>Student says, student does.</u>
        - Verify correct procedure.
    - Points of Emphasis.
        - Trimming for $V_{glide}$.
        - Wire strike avoidance.
        - Choice of landing point.
- Coach the student through a proper flow while going through the checklist (left to right!).

**Emergency Descent**

- First completion of the maneuver will be a demonstration.
    - <u>Instructor does, student says.</u>
        - Verify correct procedure.
    - Points of Emphasis.
        - Trim.
        - Area clearance.
        - $V_{FE}$.

- Second completion of the maneuver will be performed by the student.
    - Student says, student does.
        - Verify correct procedure.
    - Points of Emphasis.
        - Trim.
        - Area clearance.
        - Ensure the student does not exceed $V_{FE}$.

**Non-Towered Airport Operations**

- Proceed to Valkaria or Sebastian.
- Ensure the descent/arrival checklist is conducted.
- Guide the student through:
    - Obtaining the weather.
    - Figuring out which runway to use.
    - Determining the entry.
- The student should be making the radio calls.
    - Provide guidance as needed.
- Traffic patterns should be conducted by the student.

**Traffic Patterns**

- The student should be able to establish and execute a proper traffic pattern with minimal instructor guidance.
- It will likely still be necessary for the instructor to provide guidance during the round out and flare portions.

**Slips: Forward/Side**

- Forward:
    - Student says, student does.
        - Verify correct procedure.
        - Guard against overbanking or using too much rudder.
- Side:
    - Student says, student does.
        - Verify correct procedure.
        - The airplane must be aligned with the centerline.
        - Guard against overbanking or using too much rudder.

**Go-around/Rejected Landing**

- Have the student conduct a go-around during one of the traffic patterns.
- Ensure that the correct procedure is conducted.

**Normal and Crosswind landing**

- The student should conduct the landing.

- Provide guidance during the round and flare.
- Provide guidance on proper crosswind correction.
    - Emphasize that this must be maintained even after touchdown.

**Postflight Procedures**
- Ensure that the student conducts a thorough postflight inspection.

**Post-Brief**
- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 118.
- Assign homework:
    - Warrior Maneuver’s Manual.
        - Review non-towered operations.
        - Review steep turns.
    - Review and memorize the emergency checklists.
    - Give a copy of the completed V speed and Emergency quiz for student to learn/memorize

**Completion Standards**
This lesson is complete when the applicant can:
1. Properly identify all components of the traffic pattern.
2. Demonstrate proper sequential checklist procedures for all normal procedures without instructor guidance.
3. Conduct emergency procedures with instructor guidance.
4. Correctly perform previously learned maneuvers with minimal instructor guidance.
5. Conduct forward and sideslips with instructor guidance.
6. Make a proper go-around or continue decision with instructor guidance.
7. Conduct traffic patterns, takeoffs, and landings with instructor guidance.
8. Make radio calls with little instructor guidance.